

Assessing ESG Compliance and Impact: A Zero-Shot Learning Approach to Analyzing Fortune 500 Companies' Sustainability Reports

Armaan Agrawal
Princeton Day School
Princeton, NJ, USA
armagr25@pds.org

Abstract

In the evolving landscape of sustainable investing, environment, social, and governance (ESG) metrics are crucial for evaluating companies beyond financial performance. Recognizing the growing importance of ESG to stakeholders, companies release annual sustainability reports outlining their ESG goals and progress. This paper analyzes how Fortune 500 companies integrate ESG considerations into their operations and reporting. We extract the text from the sustainability reports, separate them into different sentences, classify them into nineteen ESG subcategories using a zero-shot learning model, and compare the determined ESG focuses to actual data to evaluate the authenticity and effectiveness of these reports. This examination unveils the current state of ESG compliance among leading corporations and provides insights into the challenges and successes of implementing sustainable practices. More importantly, this research aims to facilitate the process of analyzing lengthy and complex sustainability reports by offering a scalable and flexible approach through the use of zero-shot learning. By streamlining the analysis of these reports, this research contributes to a better understanding of corporate ESG efforts and their impact on a sustainable future.

Keywords

ESG, NLP, sustainability, zero-shot learning

1 Introduction

In the past two decades, there has been a significant shift in how companies are evaluated, with a growing emphasis on factors beyond financial performance. Among the variety of metrics that have arisen in the world of sustainable and socially-conscious investing, there are three notable ones: environment, social, and governance, collectively known as ESG.

ESG helps provide a holistic view of the structure and strategy of a corporation, playing a crucial role in responsible investing and corporate decision-making. Originating from ethical and responsible investment principles, this concept represents an enrichment and extension of corporate social responsibility (CSR), establishing itself as the core framework for the sustainable development of enterprises [1]. Each of its three components addresses distinct yet interconnected areas of corporate sustainability. The Environmental (E) aspect focuses on a company's impact on the natural environment, emphasizing the importance of resource conservation, pollution reduction, and climate change mitigation. The Social (S) component assesses how a company manages its relationships with employees, customers, suppliers, and the broader community, highlighting issues such as labor practices, human rights, and community engagement. The Governance (G) aspect examines the company's leadership,

board composition, executive compensation, and internal controls, ensuring transparency, accountability, and ethical business practices. Together, these elements provide a comprehensive framework for evaluating and enhancing a company’s long-term sustainability and social responsibility [2].

The origins of ESG can be traced back to a collaboration between the United Nations and major players in the financial industry, aiming to integrate a broader range of considerations into investment analysis and decision-making. The term ESG was coined as part of initiatives like the Global Compact and “Who Cares Wins,” which advocated for the incorporation of core values in human rights, labor standards, and environmental practices into business operations and investment decisions. These efforts emphasized the interconnectedness of ESG issues and the importance of sound corporate governance in addressing these challenges. The aim was to promote sustainable development and create more resilient financial markets by mainstreaming the integration of ESG issues into traditional investments [3].

Today, ESG factors have become increasingly important in assessing corporate value and risk, as evidenced by recent research analyzing the linkage between ESG sustainability scores and firm valuations. For instance, a study on S&P 500 firms from 2019 to 2021 showed a diminishing total ESG score, indicating that these risk factors are gaining recognition in the global economy. This shift suggests that investors are recognizing the importance of ESG criteria, as firms with strong ESG performance are often seen as better managed and more resilient. Moreover, the study found that while environmental risks did not significantly impact firm valuation, social risks negatively affected simple excess returns for both S&P 500 and Nasdaq 100 stocks [4]. These findings highlight the growing awareness among investors of the financial implications of ESG factors, including reputation, regulatory compliance, and access to capital. As a result, integrating ESG considerations into investment analysis and decision-making is becoming essential for identifying companies that are not only financially sound but also sustainable and socially responsible, ultimately contributing to a more sustainable global economy.

As ESG considerations continue to grow in importance and corporations face increasing accountability for their environmental interactions, social responsibilities, and governance structure, ESG reports have become essential in providing a comprehensive summary of a company’s ESG-related goals [5]. ESG reports help inform potential stakeholders of how a corporation’s strategy is in line with ethical standards, such as reducing carbon footprint, increasing employee diversity, or financially supporting charities [6]. However, a significant portion of these reports contain extensive content since they must address numerous different topics. This makes it more challenging to comprehensively assess the accuracy and integrity of these reports, as the large amount of content can make it difficult for stakeholders to identify key information and verify the claims made in the reports. In this vein, it is important that there exists an efficient way to verify the authenticity of these reports and ensure that there are no gaps between stated objectives and actual outcomes.

This paper provides a comprehensive analysis of how Fortune 500 companies across various industries are integrating ESG considerations into their operations and reporting. The study involves extracting and classifying statements from company sustainability reports into nineteen different ESG subcategories, allowing for a detailed comparison between stated ESG focuses and actual statistical data. This analysis not only reveals the current state of ESG compliance among leading corporations but also provides critical insights into the challenges and successes associated with implementing sustainable and socially responsible practices. By scrutinizing the gap between stated ESG objectives and actual outcomes, the paper aims to offer a clearer picture of corporate commitment to sustainability and social responsibility, ultimately contributing to a better understanding of ESG integration in corporate strategies as well as creating a more efficient pipeline to evaluate ESG investment in the corporate world. Specifically, this approach is meant to aid existing and potential investors in efficiently evaluating lengthy ESG reports in a relatively short amount of time, enabling more informed decision-making and promoting transparency and accountability in corporate sustainability practices.

2 Related Work

The integration of Natural Language Processing (NLP) techniques in the analysis of Environmental, Social, and Governance (ESG) data has seen significant advancements in recent years. Mehra et al. introduced ESGBERT, a domain-specific BERT model, pre-trained on ESG-specific corpora to enhance the classification of ESG-related texts [7]. This approach capitalizes on the contextual understanding of BERT to improve the accuracy of ESG classification tasks. Similarly, Luccioni et al. developed ClimateQA, a tool that employs a Transformer-based model for identifying climate-relevant sections in financial reports [8]. This innovation addresses the challenge of extracting actionable insights from extensive sustainability reports, facilitating more efficient analysis of climate-related disclosures.

The application of BERT-based models for multilingual ESG issue identification has also been explored. Pontes et al. investigated various strategies utilizing BERT language models for the classification of news documents across 35 distinct ESG issue labels [9]. Their study highlighted the effectiveness of the RoBERTa classifier in accurately classifying news documents in both English and French, showcasing the adaptability of BERT-based models to ESG classification tasks across different languages.

Beyond ESG analysis, NLP techniques have been employed in other financial domains as well. Nugent et al. investigated data augmentation approaches to improve the performance of NLP models in the financial sector [10]. Their research demonstrated the benefits of augmenting training data for classification tasks, highlighting the potential of data augmentation methods to enhance model robustness and generalization. In a similar context, Armbrust et al. analyzed the impact of a company’s environmental performance on its financial disclosures, illustrating the capability of NLP methods to extract information about environmental performance from financial reports [11].

These studies highlight the potential of advanced language models and NLP techniques in identifying, classifying, and analyzing ESG-related information. By leveraging the contextual understanding and adaptability of models like BERT and its variants, researchers and practitioners can enhance their ability to assess sustainability performance and inform decision-making in the context of sustainable investing and corporate governance.

Building on these advancements, this paper presents a novel approach that leverages zero-shot learning for ESG analysis. Zero-shot learning enables the classification of text into categories that were not seen during the training phase, offering a promising avenue for ESG analysis where new ESG issues and themes frequently emerge [12]. Importantly, this approach eliminates the need for a traditional training process, allowing for the direct application of the model to classify unseen categories. This feature is particularly valuable in the rapidly evolving field of ESG, where the ability to adapt to new concepts without retraining is crucial. This approach aims to further enhance the flexibility and scalability of NLP models in addressing the dynamic nature of ESG criteria, providing a more comprehensive and adaptable tool for sustainable finance research and practice.

3 Methods

3.1 Data Collection

We collected corporate sustainability reports (CSRs) from five companies across diverse industries: Apple, Walmart, Chevron, General Motors, and JP Morgan, spanning the years 2018 to 2022. These companies were specifically chosen because they represent a variety of different industries, including technology, retail, energy, automotive, and financial services. This diversity allows for a more comprehensive analysis of sustainability practices and trends across different sectors of the economy. The reports were sourced from ResponsibilityReports.com, a reputable repository for corporate responsibility and sustainability documents [13].

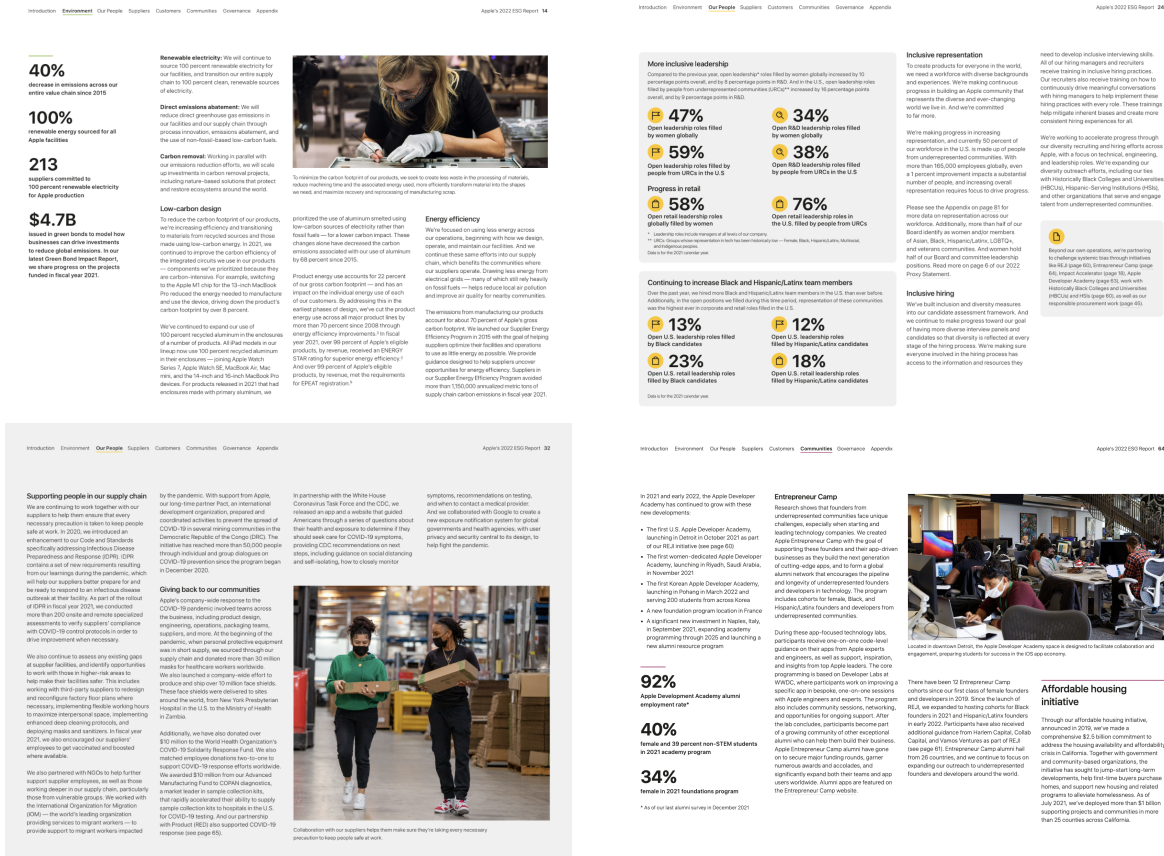


Figure 1: Apple’s 2022 Environmental, Social, and Governance (ESG) report. This report outlines Apple’s ESG goals, initiatives, and progress, highlighting the company’s commitment to sustainability and ethical business practices.

3.2 Data Preprocessing

An example report, Apple’s 2022 ESG report, is shown in Figure 1. Many of these reports are dozens of pages long, do not follow a standardized template, and contain many visual elements. Therefore, properly preprocessing the CSRs was crucial for extracting meaningful data from the complex and varied formats of the reports. To achieve this, the text preparation process was adapted from the methods detailed by Bronzini et al. [14].

The first step was to parse the PDF files of the sustainability reports using PyMuPDF and then apply text preprocessing to clean the extracted text, removing artifacts like newlines in the middle of sentences, duplicate whitespaces, and ensuring proper sentence boundaries using regular expressions [15]. After extracting the raw text, we then segment the text into individual sentences using PySBD for accurate sentence boundary disambiguation [16]. This segmentation is essential to handle the text at a granular level suitable for further analysis.

3.3 Creation of Zero-Shot Learning Model

Zero-shot learning (ZSL) models are a class of machine learning models that are trained on a set of labeled examples but then can generalize to classify new examples that belong to classes not seen during training. This capability is particularly useful in dynamic domains where new categories may emerge over time.

ID	Company	Year	Page	File Path	Statement	Label	Score	ESG
674	Apple	2022	85	/content /drive /My Drive /CSR...	In fiscal year 2021, despite the ongoing challenges presented by the COVID-19 pandemic, we interviewed 87,626 workers in our supply chain over 35,000 more than were interviewed in the previous fiscal year...	Supply Chain Management	0.89837-45575	G
675	Apple	2022	85	/content /drive /My Drive /CSR...	We also provide third-party anonymous hotlines, which are also available to our employees (see page 72), and the ability to contact the Apple Environment and...	Employee Welfare	0.80273-96798	S
676	Apple	2022	85	/content /drive /My Drive /CSR...	When an issue is raised, we require supplier management to immediately investigate and to resolve the issue in a timely manner...	Supply Chain Management	0.76441-21647	G

Table 1: Sample statements extracted from Apple’s 2022 ESG report during data preprocessing. The Zero Shot Model then classified the statements into one of the nineteen ESG subcategories, with a confidence score corresponding to each prediction.

In this project, we used Facebook’s pretrained BART-large-mnli model available on Hugging Face [17]. BART (Bidirectional and Auto-Regressive Transformers) is a transformer-based encoder-decoder (seq2seq) model that combines the strengths of BERT-like encoders and GPT-like decoders. The BART-large-mnli variant is a fine-tuned version of the BART model trained on the Multi-Genre Natural Language Inference (MultiNLI) dataset. The MultiNLI dataset consists of pairs of sentences (a hypothesis and a premise) and a label indicating whether the premise entails, contradicts, or is neutral with respect to the hypothesis [18].

We leveraged the BART-large-mnli model for zero-shot text classification by framing our classification task as a natural language inference problem. For each sentence from the corporate sustainability reports, we generated a set of hypotheses corresponding to our predefined ESG categories (e.g., “This text is about waste management”). The model then assessed the likelihood of each hypothesis being true given the input sentence [19].

This approach allowed the model to classify sentences into one of 19 ESG categories, even though it had not been explicitly trained on these categories. The ESG categories used for classification were:

- **Environmental (E):** Waste And Hazardous Materials Management, Water And Wastewater Management, Air Quality, Ecological Impacts, Energy Management, GHG Emissions.
- **Social (S):** Customer Satisfaction, Employee Engagement, Employee Welfare, Diversity and Inclusion, Human Rights, Labor Practices, Community Relations.
- **Governance (G):** Business Ethics, Critical Incident Risk Management, Product Quality And Safety, Product Design And Lifecycle Management, Supply Chain Management, Board Structure.

For each classification, the model provided a confidence score indicating the likelihood that the sentence belonged to the predicted category. These confidence scores were used to assess the model’s certainty in its predictions and to filter out less reliable classifications if needed. Table 1 shows examples of statements extracted from Apple’s 2022 ESG during the preprocessing steps, as well as the label and confidence score outputted by the model.

By utilizing the zero-shot learning capabilities of the BART-large-mnli model, we were able to dynamically categorize sentences from sustainability reports into relevant ESG categories without the need for category-specific training data. This method provided a flexible and scalable solution for analyzing the evolving content of corporate sustainability reports.

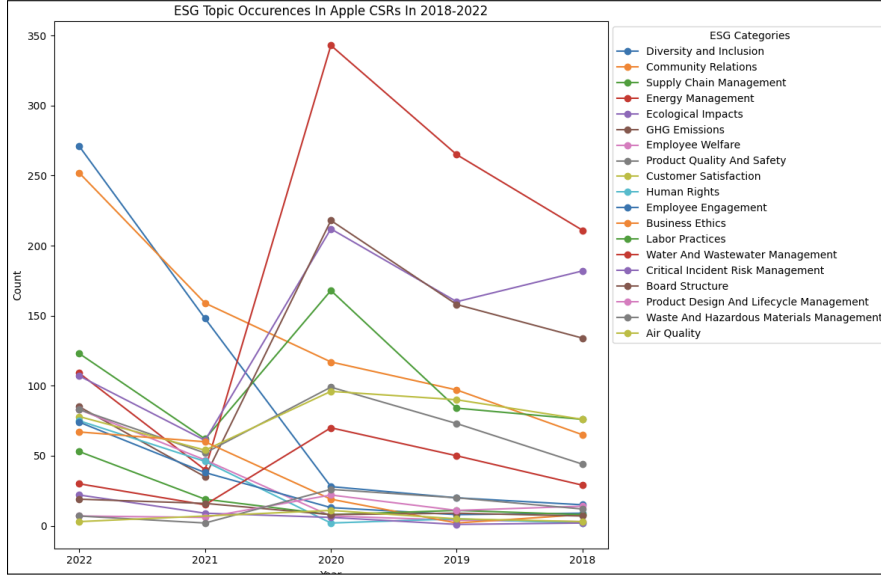


Figure 2: ESG Subcategory Evolution Plot of Apple’s corporate sustainability reports from 2018 to 2022. It provides a detailed view of the evolution of sustainability topics addressed in each report over time.

3.4 Results Visualization

To visualize the results of the ZSL model, we used the Matplotlib library to create multiple types of distribution plots for each CSR [20].

- **ESG Pillar Distribution Plot:** This plot illustrated the frequency of sentences classified into each of the three main ESG pillars: Environmental (E), Social (S), and Governance (G).
- **ESG Subcategory Distribution Plot:** This plot showed the distribution of sentences within each ESG pillar’s subcategories. For example, within the Environmental pillar, the plot displayed the frequency of sentences classified into categories such as Waste And Hazardous Materials Management, Water And Wastewater Management, etc. These plots provided a more detailed view of the specific sustainability topics addressed in each report.
- **ESG Subcategory Evolution Plot:** This plot provides a detailed view of the evolution of sustainability topics addressed in each report over time. It illustrates the distribution of sentences within each ESG pillar’s subcategories, showing how the focus areas have changed or remained consistent across different years. This plot helps identify trends and shifts in companies’ sustainability strategies, offering insights into their progress and changing priorities in environmental, social, and governance aspects. An example of this plot for Apple’s reports from 2018 to 2022 can be seen in Figure 2.

These distribution plots were instrumental in identifying trends and shifts in focus over the years, providing a visual summary of the sustainability focus areas for each company.

3.5 Analysis of Company Performance and CSR Focus

Once the most frequently mentioned categories are identified using the ZSL model, the next step involves gathering quantitative data related to these areas directly from the company’s official sources, such as annual reports, sustainability reports, or dedicated CSR sections on their websites. This data collection process ensures that the analysis is based on information provided by the company itself.

Environment						
Greenhouse gas emissions						
		Fiscal year				
		2021	2020	2019	2018	2017
Corporate emissions ¹ (metric tons CO ₂ e)	Scope 1 (gross emissions)	55,200	47,430	52,730	57,440	47,050
	Natural gas, diesel, propane	40,070	39,340	40,910	42,840	36,210
	Fleet vehicles	12,090	4,270	6,950	11,110	8,300
	Process emissions ²	3,040	3,830	4,870	3,490	2,540
	Scope 2 (market-based)	2,780	0	0	8,730	36,250
	Electricity	0	0	0	8,730	36,250
	Steam, heating, and cooling ³	2,780	-	-	-	-
	Scope 3 (gross emissions) ⁴	23,130,000	22,550,000	24,980,000	25,070,000	27,330,000
	Business travel ⁵	22,850	153,000	326,000	337,000	121,000
	Employee commute ⁶	85,570	134,000	195,000	183,000	172,000
Product life cycle emissions ⁸ (metric tons CO ₂ e)	Corporate carbon offsets ⁷	-167,000	-70,000	-	-	-
	Manufacturing (purchased goods and services)	16,200,000	16,100,000	18,900,000	18,500,000	21,100,000
	Product transportation (upstream and downstream)	1,750,000	1,800,000	1,400,000	1,300,000	1,200,000
	Product use (use of sold products)	4,990,000	4,300,000	4,100,000	4,700,000	4,700,000
	End-of-life treatment	80,000	60,000	60,000	50,000	40,000
	Product carbon offsets ⁹	-500,000	-	-	-	-
Total gross carbon footprint (without offsets) ¹⁰ (metric tons CO ₂ e)		23,200,000	22,600,000	25,100,000	25,200,000	27,500,000
Total net carbon footprint (after applying offsets) ¹⁰ (metric tons CO ₂ e)		22,530,000	22,530,000	25,100,000	25,200,000	27,500,000

Figure 3: Example of quantitative data collected from Apple’s 2022 ESG report. This data includes key performance indicators (KPIs) related to carbon emissions and footprint. Analyzing such data helps assess whether the company’s actions align with their stated ESG goals, commitments, and targets.

With the collected data in hand, a comprehensive analysis is conducted to interpret what the data signifies. This analysis involves looking for trends, changes, and patterns over time within the selected categories. For example, it might involve examining whether energy consumption has increased or decreased, whether diversity metrics have improved, or whether there have been notable shifts in greenhouse gas emissions.

One of the primary objectives of this analysis is to assess whether the company’s actual actions and initiatives align with the areas of focus identified in their CSR reports. This entails comparing the quantitative data with the stated goals, commitments, and targets outlined in the reports. The goal is to determine the extent to which the company is following through on its stated intentions.

The analysis of company performance and CSR focus over a 5-year period provides valuable insights into the companies’ long-term commitment to sustainability and their responsiveness to emerging issues. By identifying trends and shifts in focus, we can evaluate the progress made by companies in their sustainability efforts and the consistency of their actions with their reported priorities.

4 Results

Once the reports had been preprocessed, fed into the model, and visualized, clear patterns emerged on how different companies chose to disclose their sustainability practices. Thus, the statistics relevant to the focus points identified by the zero-shot classification model were gathered from the CSRs in order to explore the progress these companies made in said areas. This analysis is meant to not only highlight the diverse approaches taken by companies across different industries but also provide deeper insights into specific sustainability themes and initiatives. Refer to Section 6 in order to view more detailed figures that showcase the results.

4.1 Apple

Label	2018	2019	2020	2021	2022
Community Relations	65	97	117	159	252
Supply Chain Management	76	84	168	62	123
Customer Satisfaction	76	90	96	54	78
Diversity and Inclusion	15	20	28	148	271
Ecological Impacts	182	160	212	61	107
Energy Management	211	265	343	40	109
GHG Emissions	134	158	218	35	85
Product Quality And Safety	44	73	99	52	83
Labor Practices	8	11	8	19	53
Employee Welfare	3	4	7	47	83
Employee Engagement	9	8	13	38	74
Business Ethics	8	2	19	60	67
Water And Wastewater Management	29	50	70	15	30
Human Rights	2	5	2	46	75
Board Structure	7	9	8	16	19
Critical Incident Risk Management	2	1	6	9	22
Waste And Hazardous Materials Management	12	20	26	2	7
Product Design And Lifecycle Management	14	11	22	6	7
Air Quality	3	5	11	0	3

Table 2: Categories and their corresponding counts for the Apple CSRs from 2018 to 2022.

Apple’s approach to ESG considerations evolved significantly from 2018 to 2022, with a strong emphasis on environmental concerns during the initial years. Apple demonstrated its commitment to minimizing its environmental footprint and embracing sustainable practices across its operations, aligning with its position as a leading technology company. In terms of energy management, Apple made strides in enhancing energy efficiency and transitioning to renewable energy sources. Despite an increase in energy usage from 1,832 kWh in 2018 to 2,854 kWh in 2022, the company achieved a significant milestone by sourcing 100% of its energy from renewable sources by 2020. This shift underscores Apple’s dedication to reducing its reliance on fossil fuels and mitigating its impact on climate change. Furthermore, Apple actively worked to reduce its GHG emissions, focusing on Scope 1, Scope 2, and Scope 3 emissions, achieving a 15% reduction over five years.

However, challenges remain in Apple’s waste management and water usage practices. Hazardous waste generation increased from 3,342,700 pounds in 2018 to 6,096,600 pounds in 2020 before reducing to 3,525,840 pounds, indicating fluctuating progress in this area. Additionally, Apple’s water consumption increased by over 40%, highlighting the need for more efficient water use strategies. These metrics reveal areas where Apple still needs to improve to minimize its ecological impact fully.

In 2021 and 2022, Apple’s focus expanded to encompass social and governance aspects, reflecting the broader ESG landscape’s emphasis on these dimensions. The company shifted from releasing sustainability reports to more holistic ESG reports, signaling a pronounced commitment to transparency and accountability in social and governance matters. Apple prioritized diversity and inclusion within its workforce, making incremental progress in increasing the representation of women and individuals from diverse ethnic backgrounds. In terms of governance, Apple upheld a commitment to ethical business practices and supply chain management, implementing approximately 1,200 rigorous assessments in 2021 and 2022, as well as training programs to ensure the welfare of supplier employees and uphold human rights. Overall, Apple’s ESG journey from 2018 to 2022 highlights its comprehensive approach to addressing environmental concerns, with significant achievements in renewable energy adoption and GHG emissions reduction, while evolving its focus on social responsibility and governance to align with broader ESG trends, setting a benchmark for corporate responsibility in the tech industry [21, 22, 23, 24, 25].

4.2 Walmart

Label	2018	2019	2020	2021	2022
Community Relations	674	199	298	105	163
Supply Chain Management	432	167	255	113	107
Customer Satisfaction	290	86	142	63	83
Diversity and Inclusion	267	97	132	60	98
Ecological Impacts	256	181	252	68	76
Energy Management	189	119	139	33	30
GHG Emissions	150	84	134	44	60
Product Quality And Safety	138	40	74	11	17
Labor Practices	117	57	100	19	30
Employee Welfare	114	55	90	27	32
Employee Engagement	99	53	95	13	29
Business Ethics	90	40	92	46	27
Water And Wastewater Management	71	12	10	0	4
Human Rights	47	47	64	11	18
Board Structure	41	20	70	6	21
Critical Incident Risk Management	27	13	28	9	7
Waste And Hazardous Materials Management	19	10	21	3	2
Product Design And Lifecycle Management	9	4	4	0	0
Air Quality	6	9	7	3	1

Table 3: Categories and their corresponding counts for the Walmart CSRs from 2018 to 2022.

Walmart’s approach to ESG considerations evolved significantly from 2018 to 2022, with a notable emphasis on social responsibility and governance. The retailer demonstrated its commitment to enhancing community relations and supply chain management, aligning with its position as a global retail giant. Walmart actively engaged with local communities, with total cash and in-kind donations consistently exceeding \$1.4 billion and reaching \$1.5 billion in 2022. The increase in local cash giving from U.S. facilities, from \$42 million in 2018 to \$44 million in 2022, and the growth in participation in Walmart’s Volunteerism Always Pays (VAP) program, showcase the company’s dedication to community involvement and social impact.

Walmart also made steady progress in enhancing diversity and inclusion within its workforce. The representation of women and people of color across various organizational levels improved, with the percentage of U.S. associates who are women or people of color increasing from 55% and 44% in 2018 to 53% and 49%, respectively, in 2022. On the governance front, Walmart maintained a strong focus on supply chain management and ethical practices. The sourcing of around \$13 billion of products annually from diverse suppliers in the U.S., coupled with extensive audit assessments consistently exceeding 14,000, reinforces Walmart’s commitment to ethical supply chain practices and corporate accountability.

However, environmental concerns have not been as heavily emphasized by Walmart. While the company’s Scope 1 emissions reduced from 11.37 million metric tons CO₂e in 2018 to 9.19 million metric tons CO₂e in 2020, its Scope 2 emissions increased during the same period, from 6.15 to 7.24 million metric tons CO₂e. This indicates significant work remains regarding climate change and sustainability. In summary, Walmart’s ESG journey from 2018 to 2022 highlights its dedication to social responsibility and ethical governance. The company’s progress in aligning its operations with its ESG objectives demonstrates its commitment to driving positive change and setting an example for corporate responsibility. However, to be seen as a leader in corporate sustainability practices, Walmart should place more emphasis on improving its environmental impact. Through its continued efforts, Walmart is poised to make a significant impact on ESG in the corporate world for many years to come [26, 27, 28, 29, 30].

4.3 Chevron

Label	2018	2019	2020	2021	2022
Community Relations	118	113	423	168	194
Diversity and Inclusion	88	73	315	120	146
Energy Management	73	139	262	188	243
Ecological Impacts	55	51	265	109	144
GHG Emissions	41	69	471	302	274
Employee Engagement	29	31	72	46	51
Board Structure	24	23	82	25	39
Human Rights	23	29	62	27	23
Labor Practices	23	14	75	31	30
Water And Wastewater Management	20	20	165	53	46
Product Quality And Safety	20	24	95	23	25
Business Ethics	15	25	87	29	30
Air Quality	12	14	81	18	22
Employee Welfare	11	15	74	34	25
Customer Satisfaction	10	11	45	25	31
Critical Incident Risk Management	9	8	59	64	56
Supply Chain Management	6	5	27	27	21
Waste And Hazardous Materials Management	5	8	9	12	10
Product Design And Lifecycle Management	0	1	1	1	5

Table 4: Categories and their corresponding counts for the Chevron CSRs from 2018 to 2022

Chevron’s commitment to ESG principles from 2018 to 2022 showcases a clear emphasis on environmental responsibility, consistent with its position as one of the world’s largest oil companies. The company significantly reduced its energy consumption in operated assets and joint-venture refineries, from 940 trillion BTUs in 2018 to 784 trillion BTUs in 2022. Chevron also made commendable progress in reducing its Scope 1 emissions, including direct operated emissions (from 68 million metric tons COe in 2018 to 53 million metric tons COe in 2022) and direct equity emissions (from 66 million metric tons COe in 2018 to 53 million metric tons COe in 2022). However, Scope 2 and 3 emissions remained relatively consistent, indicating areas where Chevron could focus more efforts.

In contrast to its environmental achievements, Chevron’s social responsibility initiatives, including Diversity, Equity, and Inclusion (DEI) and community support, demonstrated less prominence between 2018 and 2022. The composition of the workforce showed limited change, with women remaining at 30% and ethnic minorities increasing slightly from 41% to 43%. Contributions to charitable organizations and nonprofits were steady but not as substantial as in other ESG areas, suggesting room for growth in social and community-focused initiatives.

In summary, Chevron’s ESG efforts from 2018 to 2022 reflect a strong dedication to environmental sustainability, with significant reductions in energy consumption and direct GHG emissions. However, the company’s social responsibility initiatives have shown less prominence, indicating an opportunity for Chevron to further strengthen its commitment to social and community-focused initiatives. By enhancing workforce diversity and increasing contributions to charitable organizations, Chevron can achieve a more balanced and comprehensive approach to ESG, positioning itself as a leader in corporate responsibility within the energy sector [31, 32, 33, 34, 35].

4.4 General Motors

Label	2018	2019	2020	2021	2022
Community Relations	394	591	352	207	198
Energy Management	351	456	475	265	394
Product Quality And Safety	249	244	230	149	142
Diversity and Inclusion	235	231	324	208	181
GHG Emissions	199	335	248	165	135
Supply Chain Management	197	172	215	173	149
Customer Satisfaction	178	209	296	121	99
Ecological Impacts	164	230	408	91	102
Employee Engagement	128	120	185	90	68
Business Ethics	80	77	81	95	83
Labor Practices	74	114	112	45	41
Employee Welfare	60	58	74	49	29
Board Structure	60	64	78	40	43
Water And Wastewater Management	48	93	111	21	52
Critical Incident Risk Management	33	34	48	19	33
Human Rights	27	54	82	66	65
Air Quality	21	58	41	5	9
Product Design And Lifecycle Management	19	14	20	5	10
Waste And Hazardous Materials Management	16	29	36	4	6

Table 5: Categories and their corresponding counts for General Motors from 2018-2022.

General Motors (GM) has demonstrated a multifaceted approach to Environmental, Social, and Governance (ESG) principles from 2018 to 2022, aligning with its position as a leading automotive manufacturer. In the environmental sphere, GM has made significant strides in reducing its ecological footprint. The company worked diligently to decrease the energy used in its operations per vehicle, from 2.13 MWh in 2018 to 2.30 MWh in 2022, with some fluctuations. Despite this, GM’s commitment to renewable energy sources is evident, with renewables comprising only 3% of the company’s global electricity use in 2018, jumping to 30% by 2022. This transition highlights GM’s dedication to sustainable energy practices and its efforts to mitigate environmental impact.

In terms of social responsibility, GM has maintained a focus on diversity and inclusion within its workforce. The company has consistently balanced the distribution of men and women, with around 55% women and 45% men. Furthermore, GM’s workforce is notably diverse, with 47% of employees being people of color (POC) among the 87% who reported their ethnicity. This indicates GM’s growing commitment to fostering a diverse and inclusive work environment. Additionally, GM has significantly boosted its community engagement efforts, particularly through grants provided to non-profit and charitable organizations. In 2018, GM provided \$3.2 million in grants, which fluctuated over the years, reaching a remarkable \$60 million in 2022, signifying a substantial commitment to community relations.

In conclusion, GM’s ESG commitment from 2018 to 2022 highlights its proactive approach to environmental sustainability, an evolving commitment to diversity and inclusion, and a significant increase in community engagement efforts. These efforts underscore GM’s dedication to making a positive impact on the environment, its workforce, and the communities it serves, demonstrating its comprehensive commitment to ESG principles. Through these continued efforts, GM is poised to set a benchmark for corporate responsibility in the automotive industry [36, 37, 38, 39, 40].

4.5 JP Morgan

Label	2018	2019	2020	2021	2022
Community Relations	191	193	179	577	546
Diversity and Inclusion	134	127	226	460	433
Customer Satisfaction	131	125	74	111	103
Ecological Impacts	78	38	52	246	264
Energy Management	70	53	107	243	197
Employee Welfare	55	59	28	41	44
Employee Engagement	54	44	32	92	84
Business Ethics	53	51	63	163	152
Product Quality And Safety	35	32	5	26	11
GHG Emissions	22	69	167	96	102
Board Structure	20	20	30	182	178
Labor Practices	14	11	11	24	52
Human Rights	13	7	13	58	73
Critical Incident Risk Management	12	11	34	27	36
Water And Wastewater Management	9	2	6	12	16
Supply Chain Management	5	1	8	16	19
Air Quality	5	2	4	7	8
Waste And Hazardous Materials Management	1	1	2	5	1
Product Design And Lifecycle Management	0	1	0	3	4

Table 6: Categories and their corresponding counts for JP Morgan from 2018-2022.

JP Morgan Chase (JP Morgan) has undertaken a comprehensive journey in embracing Environmental, Social, and Governance (ESG) principles from 2018 to 2022, highlighting its commitment to environmental sustainability, diversity and inclusion, and community engagement. In the environmental domain, JP Morgan successfully reduced its total energy consumption from 2,615,787 MWh in 2018 to 2,463,279 MWh by 2022. Additionally, the bank reduced its Scope 1 GHG emissions from 102,423 metric tons COe in 2018 to 88,553 metric tons COe in 2022, and Scope 2 emissions from 851,622 metric tons COe in 2018 to 783,616 metric tons COe in 2022. These reductions reflect JP Morgan’s efforts to optimize energy usage, minimize its carbon footprint, and contribute to global climate change mitigation.

In terms of social responsibility, JP Morgan maintained gender diversity within its workforce, with a balanced distribution of 49% women and 51% men. The bank also made efforts to diversify its leadership roles, indicating ongoing initiatives to enhance gender diversity at the leadership level. Additionally, JP Morgan’s workforce displayed increasing ethnic diversity, with the percentage of white employees decreasing from 52% in 2020 to 44% in 2022, and an increase in Hispanic, Asian, and Black employees. This shift underscores the bank’s commitment to fostering a diverse and inclusive workplace.

JP Morgan’s commitment to community engagement is evident through its substantial investments. By 2022, the bank had spent nearly \$18 billion on building affordable housing, \$1.1 billion on loans to businesses owned by people of color, and \$1.8 billion on low-income housing credit. These initiatives highlight JP Morgan’s dedication to making a positive impact in the communities it serves and its commitment to ESG values. In conclusion, JP Morgan’s ESG efforts from 2018 to 2022 demonstrate a proactive approach to environmental sustainability, diversity and inclusion, and community engagement. Through its continued efforts, JP Morgan sets a benchmark for corporate responsibility in the financial sector [41, 42, 43, 44, 45]

5 Conclusion and Future Work

Through the analysis of the ESG commitments of Walmart, Chevron, General Motors, and JP Morgan Chase from 2018 to 2022, we have observed several general insights and trends regarding how these companies abide by their stated ESG goals.

Firstly, there is a clear emphasis on environmental sustainability across all four companies, with significant efforts directed towards reducing energy consumption and greenhouse gas emissions. This aligns with the global push toward mitigating climate change and highlights the companies' commitment to environmental stewardship.

Secondly, diversity and inclusion emerge as important themes, with varying degrees of progress among the companies. While some have made notable strides in enhancing workforce diversity and fostering inclusive work environments, others show room for improvement. This indicates that while there is a general acknowledgment of the importance of social responsibility, the degree of commitment and implementation varies.

Thirdly, community engagement remains a consistent aspect of the companies' ESG goals, with each demonstrating a commitment to contributing positively to the communities they operate in. This is reflected in their charitable donations, volunteer programs, and community development initiatives.

One of the key takeaways from this project is the efficiency of the method of analysis employed. By categorizing and quantifying ESG-related activities and achievements, we were able to quickly gain insights into the companies' alignment with their ESG goals. This rapid assessment approach allows for a more dynamic and responsive evaluation of ESG performance, which is crucial in the fast-evolving landscape of corporate sustainability. Using this model, stakeholders can quickly identify what a company is focusing on in its ESG efforts and then assess that focus without having to analyze a report that is dozens of pages long. This streamlined approach enables a more efficient and targeted evaluation of a company's ESG commitments.

In terms of general insights, it appears that companies are increasingly aligning their operations with their stated ESG goals, particularly in the environmental domain. However, the degree of alignment varies across different ESG aspects, with some companies showing stronger commitment and progress in certain areas than others. This variability highlights the complexity of ESG implementation and the need for ongoing monitoring and assessment.

Looking forward, there is significant potential for future work in this area. Comparing the insights gained from this analysis to existing ESG scores could provide valuable perspectives on their correlation and reliability. Moreover, this research could serve as a foundation for developing new methodologies to calculate ESG scores that more accurately reflect a company's actual performance and impact. By incorporating a broader range of data and employing more nuanced analytical techniques, we can enhance the effectiveness of ESG assessments. This, in turn, could contribute to more informed decision-making by companies and stakeholders, and better alignment with sustainability goals.

References

- [1] Guochao Wan and et al. "Hotspots and Trends of Environmental, Social and Governance (ESG) Research: A Bibliometric Analysis". In: *Data Science and Management* 6.2 (2023), pp. 65–75. DOI: [10.1016/j.dsm.2023.03.001](https://doi.org/10.1016/j.dsm.2023.03.001). URL: <https://doi.org/10.1016/j.dsm.2023.03.001>.
- [2] H. Ahmad, M. Yaqub, and S.H. Lee. "Environmental-, social-, and governance-related factors for business investment and sustainability: a scientometric review of global trends". In: *Environ Dev Sustain* 26 (2024), pp. 2965–2987. DOI: [10.1007/s10668-023-02921-x](https://doi.org/10.1007/s10668-023-02921-x). URL: <https://doi.org/10.1007/s10668-023-02921-x>.

- [3] Elizabeth Pollman. “The Making and Meaning of ESG”. In: *U of Penn, Inst for Law Econ Research Paper No. 22-23, European Corporate Governance Institute - Law Working Paper No. 659/2022, Harvard Business Law Review, Forthcoming* (2022). URL: <https://ssrn.com/abstract=4219857>.
- [4] G. Cohen. “The impact of ESG risks on corporate value”. In: *Rev Quant Finan Acc* 60 (2023), pp. 1451–1468. DOI: [10.1007/s11156-023-01135-6](https://doi.org/10.1007/s11156-023-01135-6). URL: <https://doi.org/10.1007/s11156-023-01135-6>.
- [5] C. S. D. S. Lokuwaduge and K. Heenetigala. “Integrating Environmental, Social and Governance (ESG) Disclosure for a Sustainable Development: An Australian Study”. In: *Bus. Strat. Env.* 26 (2017), pp. 438–450. DOI: [10.1002/bse.1927](https://doi.org/10.1002/bse.1927).
- [6] Guido Giese et al. “Foundations of ESG Investing: How ESG Affects Equity Valuation, Risk, and Performance”. In: *The Journal of Portfolio Management* 45 (2019), pp. 69–83. DOI: [10.3905/jpm.2019.45.5.069](https://doi.org/10.3905/jpm.2019.45.5.069).
- [7] S. Mehra, R. Louka, and Y. Zhang. “ESGBERT: Language Model to Help with Classification Tasks Related to Companies’ Environmental Social and Governance Practices”. In: *Proceedings of the International Conference on Computer Science and Information Technology*. 2022.
- [8] A. Luccioni, E. Baylor, and N. Duchene. “Analyzing Sustainability Reports Using Natural Language Processing”. In: *arXiv preprint arXiv:2011.08073* (2020).
- [9] E. L. Pontes, M. Benjannet, and L. K. Ming. “Leveraging BERT Language Models for Multi-Lingual ESG Issue Identification”. In: *arXiv preprint arXiv:2309.02189* (2023).
- [10] T. Nugent, N. Stelea, and J. L. Leidner. “Detecting ESG topics using domain-specific language models and data augmentation approaches”. In: *arXiv preprint arXiv:2010.08319* (2020).
- [11] F. Armbrust, H. Schäfer, and R. Klinger. “A Computational Analysis of Financial and Environmental Narratives within Financial Reports and its Value for Investors”. In: *Proceedings of the 1st Joint Workshop on Financial Narrative Processing and MultiLing Financial Summarisation*. 2020.
- [12] Y. Xian et al. “Zero-Shot Learning – A Comprehensive Evaluation of the Good, the Bad and the Ugly”. In: *arXiv* (2017). URL: <https://doi.org/10.48550/ARXIV.1707.00600>.
- [13] *responsibilityreports.com*.
- [14] M. Bronzini, C. Nicolini, and B. Lepri. “Glitter or gold? Deriving structured insights from sustainability reports via large language models”. In: *EPJ Data Sci.* 13 (2024), p. 41. DOI: [10.1140/epjds/s13688-024-00481-2](https://doi.org/10.1140/epjds/s13688-024-00481-2). URL: <https://doi.org/10.1140/epjds/s13688-024-00481-2>.
- [15] URL: <https://pypi.org/project/PyMuPDF>.
- [16] URL: <https://pypi.org/project/pysbd>.
- [17] URL: <https://huggingface.co/facebook/bart-large-mnli>.
- [18] Adina Williams, Nikita Nangia, and Samuel Bowman. “A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference”. In: *Proceedings of the NAACL*. 2018, pp. 1112–1122. DOI: [10.18653/v1/N18-1101](https://doi.org/10.18653/v1/N18-1101).
- [19] W. Yin, J. Hay, and D. Roth. “Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach”. In: *arXiv* (2019). URL: <https://doi.org/10.48550/ARXIV.1909.00161>.
- [20] URL: <https://matplotlib.org>.
- [21] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReports/PDF/NASDAQ_AAPL_2022.pdf.
- [22] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/a/NASDAQ_AAPL_2021.pdf.
- [23] URL: https://www.apple.com/environment/pdf/Apple_Environmental_Progress_Report_2020.pdf.
- [24] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/a/NASDAQ_AAPL_2019.pdf.

- [25] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/a/NASDAQ_AAPL_2018.pdf.
- [26] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/w/NYSE_WMT_2022.pdf.
- [27] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/w/NYSE_WMT_2021_3036d8b69ebe45ae9bc770a7b99a8a8b.pdf.
- [28] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/w/NYSE_WMT_2020_68e82aaa21eb452ea09e7e5f53d0472f.pdf.
- [29] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/w/NYSE_WMT_2019.pdf.
- [30] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/w/NYSE_WMT_2018_093758797bbc4cdea3f1260f230776b9.pdf.
- [31] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReports/PDF/NYSE_CVX_2022.pdf.
- [32] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/c/NYSE_CVX_2021.pdf.
- [33] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/c/NYSE_CVX_2020.pdf.
- [34] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/c/NYSE_CVX_2019.pdf.
- [35] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/w/NYSE_WMT_2018_093758797bbc4cdea3f1260f230776b9.pdf.
- [36] URL: https://www.gmsustainability.com/_pdf/resources-and-downloads/GM_2022_SR.pdf.
- [37] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/g/NYSE_GM_2021.pdf.
- [38] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/g/NYSE_GM_2020.pdf.
- [39] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/g/NYSE_GM_2019.pdf.
- [40] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/g/NYSE_GM_2018.pdf.
- [41] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReports/PDF/NYSE_JPM_2022.pdf.
- [42] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/j/NYSE_JPM_2021.pdf.
- [43] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/j/NYSE_JPM_2020.pdf.
- [44] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/j/NYSE_JPM_2019.pdf.
- [45] URL: https://www.responsibilityreports.com/HostedData/ResponsibilityReportArchive/j/NYSE_JPM_2018.pdf.

6 Supplementary Information

The following figures visually represent the ESG subtopic distribution for each of the five companies' reports from 2018 to 2022. The first represents the topic distribution year-by-year, and the second shows the evolution of the frequencies of each of the topics. Please note that the index values in the first figure for each company map to the ESG subtopics according to the below table.

Category	Index
Waste And Hazardous Materials Management	0
Water And Wastewater Management	1
Air Quality	2
Ecological Impacts	3
Energy Management	4
GHG Emissions	5
Customer Satisfaction	6
Employee Engagement	7
Employee Welfare	8
Diversity and Inclusion	9
Human Rights	10
Labor Practices	11
Community Relations	12
Business Ethics	13
Critical Incident Risk Management	14
Product Quality And Safety	15
Product Design And Lifecycle Management	16
Supply Chain Management	17
Board Structure	18

Table 1: ESG Subtopic Index Mapping

6.1 Apple

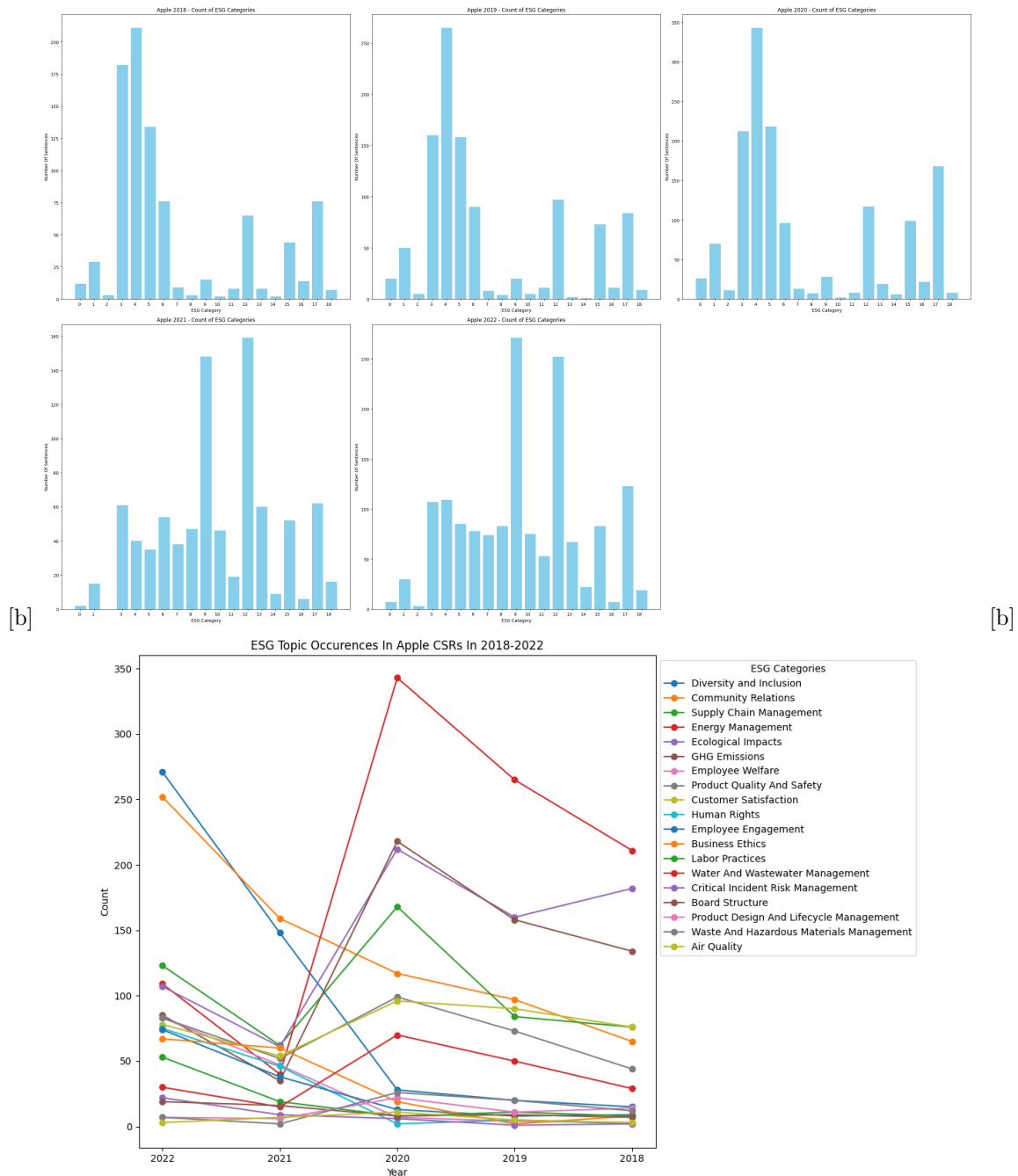


Figure A: Apple 2018-2022 Count and Evolution of ESG Categories

6.2 Walmart

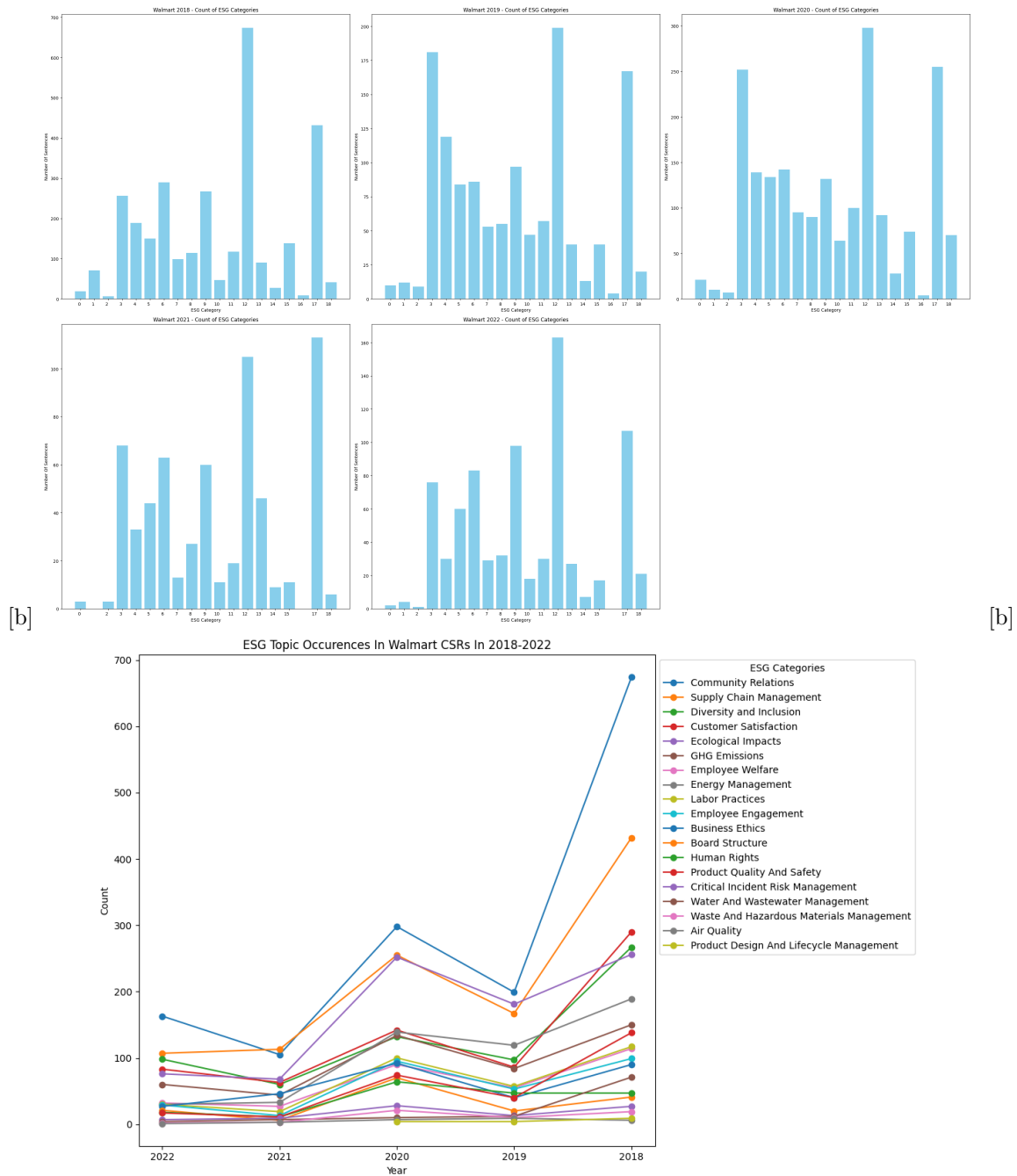


Figure B: Walmart 2018-2022 Count and Evolution of ESG Categories

6.3 Chevron

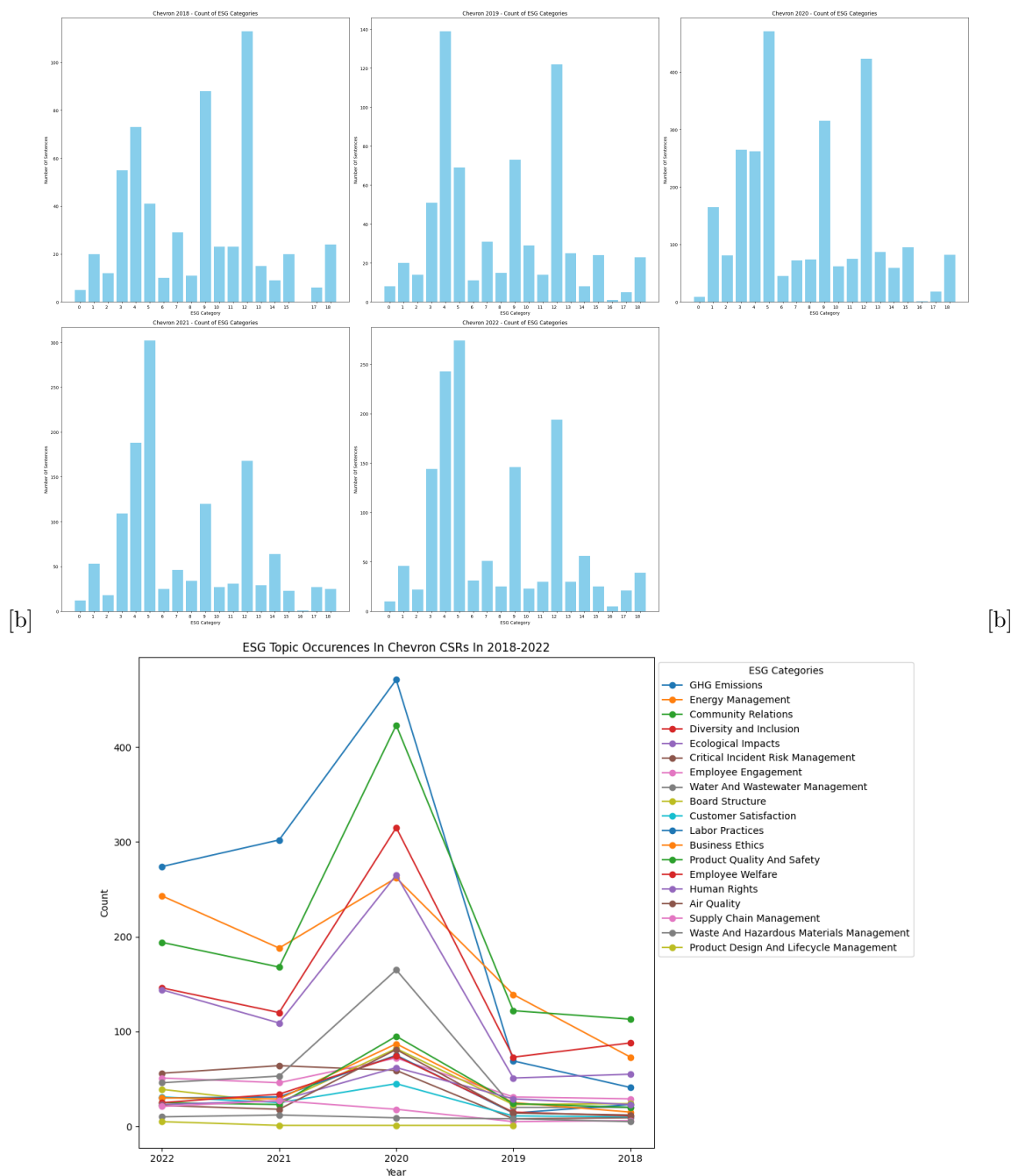


Figure C: Chevron 2018-2022 Count and Evolution of ESG Categories

6.4 General Motors

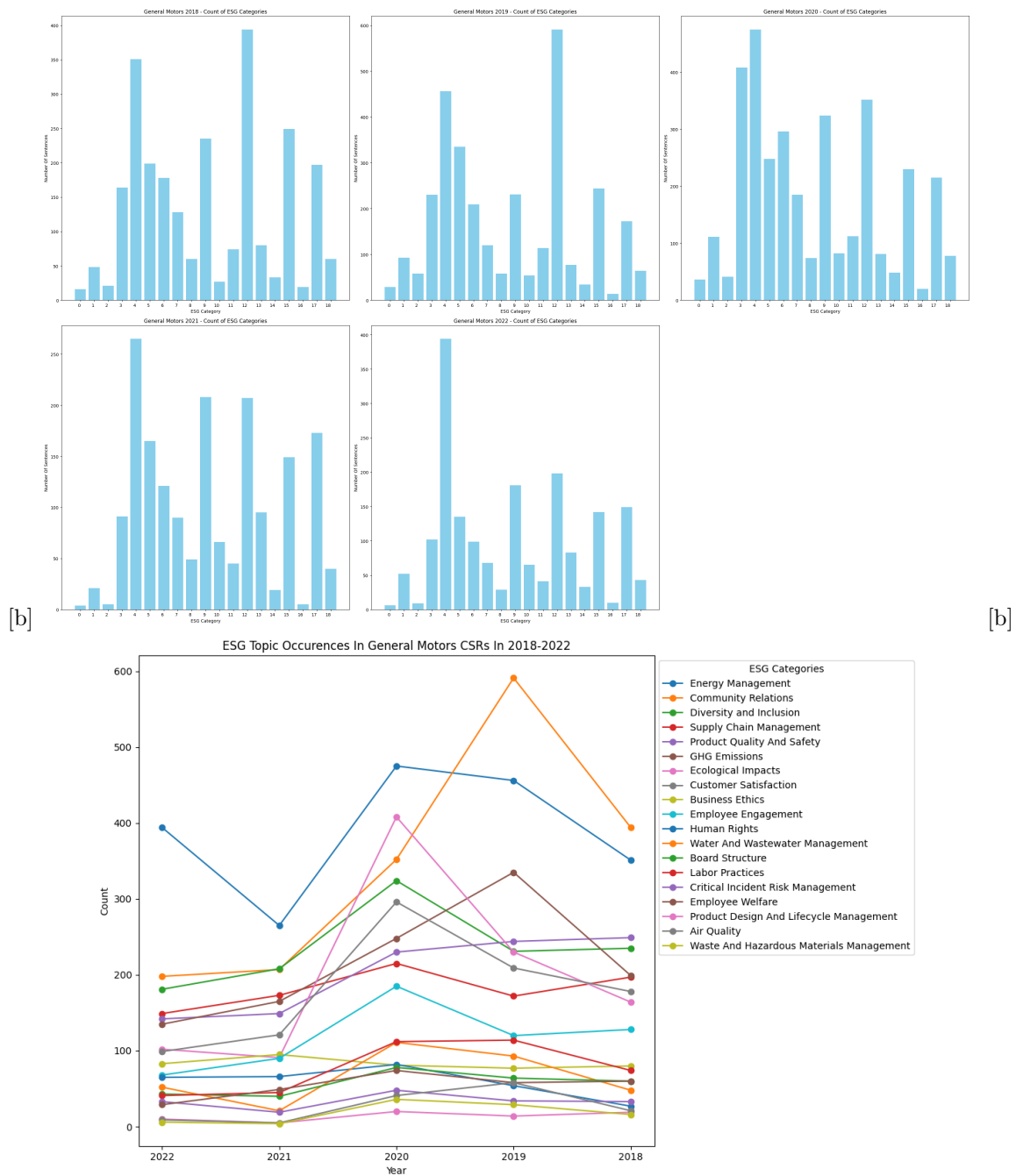


Figure D: General Motors 2018-2022 Count and Evolution of ESG Categories

6.5 JP Morgan

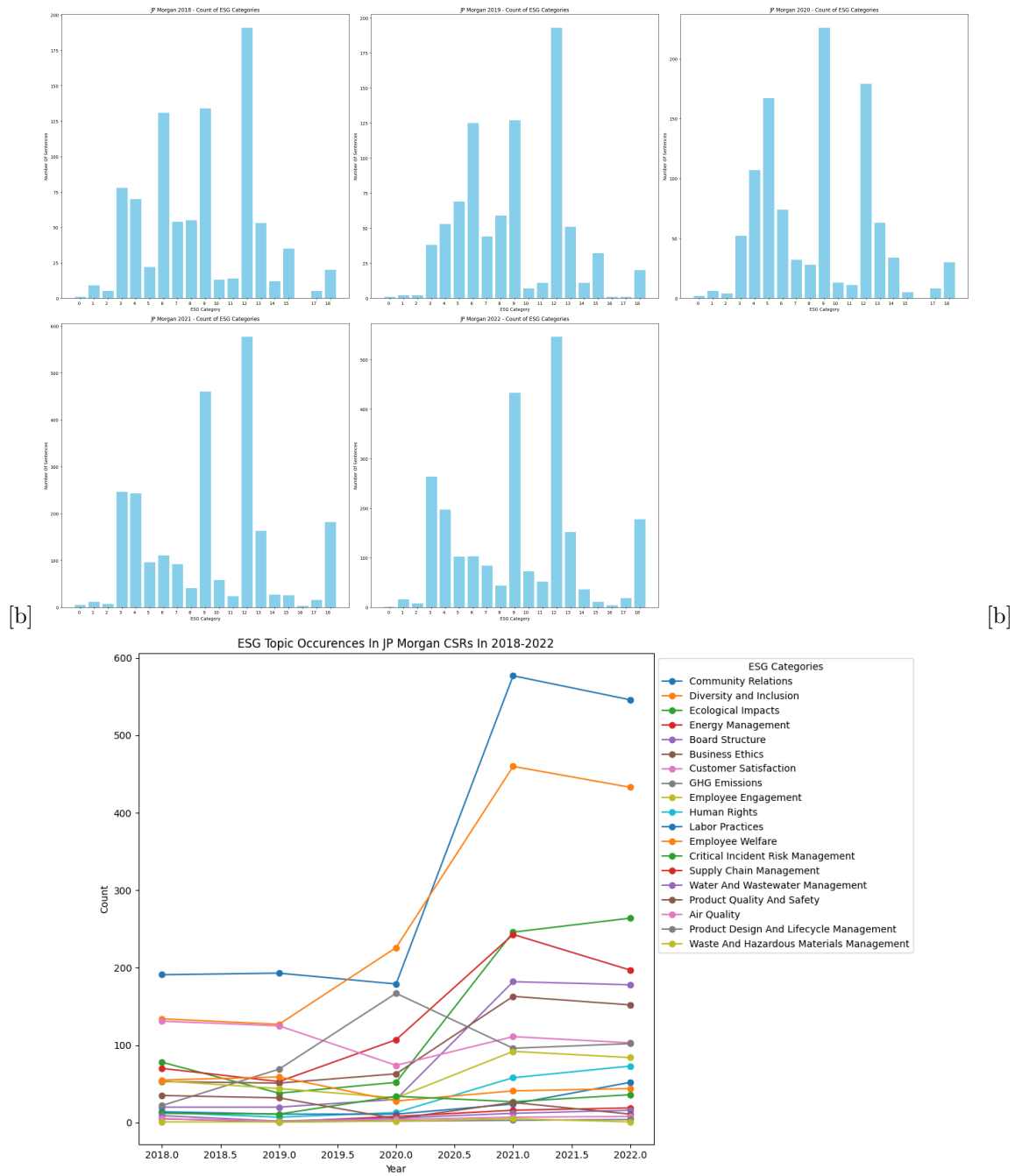


Figure E: JP Morgan 2018-2022 Count and Evolution of ESG Categories